MA2104(3)

Test 1

Wednesday, 09/09/2025 Time: 45 minutes

1. (8 points) Determine all $z \in \mathbb{C}$ such that $\tanh z = \frac{i}{2}$

2. (a) (3 points) Prove that for $z \in \mathbb{C}$,

$$\overline{\cos z} = \cos \overline{z}$$

(b) (7 points) Find, in the form $re^{i\theta}$, all the roots of the equation

$$z^6 - z^3 + 1 = 0$$

3. (7 points) Use De Moivre's theorem to evaluate

$$\int_0^{\frac{\pi}{2}} \cos^3\theta \sin^2\theta \, \mathrm{d}\theta$$

Total: 25 marks